

100 Days of Azure DevOps

Day 25 Handout — npm ci + Cache@2 — stop paying hosted minutes to watch paint dry

3 - Continuous Integration · Personal lab only · Educational content · Not a sales pitch

Architecture A — Node CI path (lockfile is the contract)

Step	Task / command	Why
Runtime pin	NodeTool@0 versionSpec 20.x	Not whatever ubuntu-latest shipped this week
Cache key	Cache@2 key: npm \$(Agent.OS) package-lock.json	Restore node_modules when lockfile unchanged
Install	npm ci (not npm install)	Fails if lockfile is out of sync — that is a feature
Test	npm test	Even a one-line Node assert counts
Optional build	npm run build if you have it	Skip if the lab app is a script
Lockfile	package-lock.json committed	Cache@2 key is useless without it
Path trigger	src/sample-node/**	Docs-only commits should not install npm

Architecture B — npm ci vs install vs cache

Choice	Lab default	Avoid when
npm ci	Always in CI	You do not have package-lock.json yet — generate and commit it
npm install	Local laptop only	CI: it rewrites the lockfile and hides drift
Cache@2 path node_modules	Yes, with lockfile in the key	You cache ~/.npm but still run npm install without ci
Node 20.x	LTS pin	Floating node: latest
Private feed	Later: .npmrc + npmAuthenticate@0 to day09-packages	PAT in .npmrc committed
windows-latest	Only if native modules need MSVC	Default ubuntu-latest

Architecture C — pitfalls that miss the cache or ship node_modules

Pitfall	What you see	Fix
Cache key without lockfile hash	Cache never hits, or hits the wrong tree	Include package-lock.json in Cache@2 key
npm install in CI	lockfile dirty; 'works on my agent'	npm ci; commit the lockfile
Committed node_modules	PR is 10k files; cache pointless	gitignore + git rm -r --cached node_modules
No restoreKeys	First build after tiny lock change is cold	restoreKeys: npm \$(Agent.OS)
Scripts.test missing	npm ERR! missing script: test	Add test script even if it is node --test
Cache@2 path wrong	cache miss every run	path: src/sample-node/node_modules matching cd

One-liner to remember

Node CI without caching is watching paint dry while paying Microsoft-hosted minutes for the privilege.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal azure-100-labs. Tiny app under /src/sample-node. Commit package-lock.json. Hosted ubuntu-latest. No work npm registries.

- 1 mkdir src/sample-node && create package.json with name sample-node, scripts.test running node --test or a one-liner. npm install locally to produce package-lock.json.
- 2 Confirm .gitignore has node_modules/. Commit package.json + package-lock.json only on feature/day25-node-ci.
- 3 Add pipelines/node-ci.yml with NodeTool@0, Cache@2, npm ci, npm test.
- 4 Create a pipeline pointing at pipelines/node-ci.yml. Run it twice: first run may be a cache miss; second should hit Cache@2 if lockfile unchanged.
- 5 Open the Cache@2 log: look for Cache restored from / Cache saved. That is the receipt.
- 6 PR + squash merge. Do not put a PAT in .npmrc.
- 7 Optional: Connect to feed day09-packages later — not required for the hello test.

pipelines/node-ci.yml — Node 20, Cache@2, npm ci, npm test

```
trigger:
  branches:
    include:
      - main
  paths:
    include:
      - src/sample-node/**
      - pipelines/node-ci.yml
pr:
  branches:
    include:
      - main

pool:
  vmImage: ubuntu-latest

steps:
- task: NodeTool@0
  inputs:
    versionSpec: 20.x

- task: Cache@2
  inputs:
    key: 'npm | "$(Agent.OS)" | src/sample-node/package-lock.json'
    restoreKeys: |
      npm | "$(Agent.OS)"
    path: src/sample-node/node_modules

- script: |
  cd src/sample-node
  npm ci
  npm test
  displayName: npm ci and test
```

Done checklist

- src/sample-node has package.json + package-lock.json; node_modules is gitignored
- Pipeline uses NodeTool@0 20.x and npm ci (not npm install)
- Cache@2 key includes the lockfile; second run can restore
- npm test is green on ubuntu-latest

Tomorrow — Day 26

CI for Python

